

Full Proofs: Supplementary Derivations for *Signed Decomposition of the Regression Structure via Linear JEPA Training*

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Abstract

This document supplements the paper [1], AIS 2026 camera-ready. Five results there are stated with a proof sketch rather than the full derivation, for page-budget reasons (a hard 15-page cap). This document carries the full, previously-verified derivation each sketch is compressed from, unabridged. Nothing here is new mathematics — every derivation below already existed in a prior draft of the paper and was moved here, not re-derived, when the paper’s own proof was shortened to a sketch. Each section is headed by the exact label the corresponding result carries in the main paper’s L^AT_EX source, so a reader can match sketch to full proof unambiguously regardless of section/theorem numbering in either document. Notation matches the main paper throughout; see its §1 (Notation) for definitions not restated here.

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1 Generalised diagonal ODE (prop:diagonal-ode)

Proposition 1.1 (Generalised diagonal ODE). *Let $\sigma_r(t)$ be the diagonal amplitudes of the paper’s Definition 1 (signed generalised eigenbasis). Under the paper’s Proposition 2 (quasi-static decoder, rigorous) and Assumption 3 (strict spectral gap),*

$$\dot{\sigma}_r(t) = L \mu_r \sigma_r(t)^{2-1/L} (\rho_r^* - \sigma_r(t)^L) + R_r(t, \varepsilon), \quad |R_r(t, \varepsilon)| \leq C_R \varepsilon^{(2L-1)/L}, \quad (1)$$

uniformly on $[0, t_{\max}^]$, with C_R depending on the population covariances, L , and the spectral gap δ but not on ε .*

Proof. The first step is the idealisation the main paper’s Proof strategy flags as the one place the whole argument leans on classical, not re-derived, machinery: treating $\bar{W}(t)$ as exactly diagonal in

the eigenbasis (the correction being the off-diagonal bound of `lem:offdiag`) lets v_r^* stand in for a two-sided eigenvector of $\bar{W}(t)\bar{W}(t)^\top$ and $\bar{W}(t)^\top\bar{W}(t)$, so the preconditioned flow projects onto

$$\dot{\sigma}_r(t) \approx v_r^{*\top} \dot{\bar{W}}(t) v_r^* = -P_{rr}(t) \cdot v_r^{*\top} \nabla_{\bar{W}} \mathcal{L}(\bar{W}(t), V(t)) v_r^* = -L \sigma_r(t)^{2(L-1)/L} \cdot v_r^{*\top} \nabla_{\bar{W}} \mathcal{L} v_r^*,$$

using the preconditioner formula $P_{rr}(t) = L \sigma_r(t)^{2(L-1)/L}$ (the paper's Lemma, coupled gradient-flow system); we do not separately verify v_r^* 's two-sidedness here beyond the diagonality idealisation itself.

The second step, by contrast, is exact given that idealisation, not an approximation on top of it. Write $\Delta V(t) := V(t) - V_{\text{qs}}(\bar{W}(t))$. Since $\bar{W} \Sigma^{xx} \bar{W}^\top$ is invertible (the paper's Corollary, PD lower bound), $V_{\text{qs}}(\bar{W}) \bar{W} \Sigma^{xx} = \bar{W} \Sigma^{yx}$ exactly, so substituting $V = V_{\text{qs}} + \Delta V$ into the gradient formulas and using the eigenpair relation $\Sigma^{yx} v_r^* = \rho_r^* \Sigma^{xx} v_r^*$ collapses every V_{qs} -only term:

$$v_r^{*\top} \nabla_{\bar{W}} \mathcal{L} v_r^* = (V v_r^*)^\top (V - \rho_r^* I) \bar{W} \Sigma^{xx} v_r^* = (V v_r^*)^\top \Delta V \bar{W} \Sigma^{xx} v_r^*,$$

an *exact* identity, not an approximation. Two things follow. First, this is why the substitution is called *quasi-static*: the entire driving term is proportional to the tracking error ΔV , so the leading-order Bernoulli drift $L \mu_r \sigma_r^{2-1/L} (\rho_r^* - \sigma_r^L)$ cannot come from bounding $\|\Delta V\|$ as an independent small quantity via the quasi-static Proposition alone — doing so, via Cauchy–Schwarz, gives only $\dot{\sigma}_r = O(\sigma_r^{2(L-1)/L} \varepsilon^{2(L-1)/L})$, one order coarser in ε than the $\varepsilon^{(2L-1)/L}$ claimed above. The leading term instead requires ΔV 's own self-consistent leading-order value under the coupled $(\bar{W}, V)$ dynamics — a singular-perturbation reduction, not a plug-in bound. Second, the residual R_r is therefore exactly two contributions, not three: (i) the gap between ΔV 's crude quasi-static-Proposition bound and its true self-consistent size, and (ii) the off-diagonal correction to the diagonality idealisation used above, controlled by the off-diagonal bound. (The gradient formulas contain no Σ^{yy} term, so no third, independent contribution from Σ^{yy} arises.) $\square$

Lean status (see main paper Appendix A / this repository's crosswalk). `Saxe.diagAmp_ODE` is sorry-free but the constant C_R above is currently pinned via a compactness argument rather than a proof that it is genuinely independent of ε — a real, disclosed open gap, unaffected by the relocation of this proof.

2 Plateau estimator (prop:plateau)

Proposition 2.1 (Plateau estimator). *For $\rho_r^* > 0$, the plateau estimator $\hat{\rho}_r^{\text{plateau}}(\varepsilon, T) := \sigma_r(T)^L$ satisfies*

$$|\hat{\rho}_r^{\text{plateau}}(\varepsilon, T) - \rho_r^*| \leq C_p \cdot \varepsilon^{1/L} \cdot |\log \varepsilon| \quad \forall \varepsilon < \varepsilon_0, \forall T \geq T_*(\rho_r^*, \mu_r, \varepsilon), \quad (2)$$

where T_* is the trajectory-derived plateau-detection time (definable from σ_r alone, no parameter input) and the constants ε_0, C_p, T_* depend on L, ρ_r^*, μ_r , and the residual constant C_R of §1.

Proof. Write $u(t) := \sigma_r(t) - (\rho_r^*)^{1/L}$. Linearising the bracket $\rho_r^* - \sigma_r^L$ at $\sigma_r = (\rho_r^*)^{1/L}$ gives $\rho_r^* - \sigma_r^L = -L(\rho_r^*)^{(L-1)/L} u + O(u^2)$, so (1) becomes, for u small,

$$\dot{u}(t) = -c u(t) + R_r(t, \varepsilon) + O(u(t)^2), \quad c := L^2 \mu_r (\rho_r^*)^{(2L-2)/L} > 0,$$

a constant (in ε) contraction rate. Dropping the quadratic remainder (justified once u is confined to a fixed neighbourhood of 0, since there it is dominated by the linear term) and applying the variation-of-constants formula to the resulting linear inequality gives, for all $t \in [0, t_{\max}]$,

$$|u(t)| \leq |u(0)| e^{-ct} + \frac{C_R}{c} \varepsilon^{(2L-1)/L} (1 - e^{-ct}) \leq |u(0)| e^{-ct} + \frac{C_R}{c} \varepsilon^{(2L-1)/L}.$$

Choose $T_*(\varepsilon)$ so that the exponential transient itself has decayed to the target order, $|u(0)|e^{-cT_*} = \varepsilon^{1/L}|\log \varepsilon|$: since $|u(0)| = O(1)$, this gives $T_* = \Theta(|\log \varepsilon|)$ — the trajectory-derived plateau time, definable from σ_r alone since it only requires detecting when σ_r 's rate of change has fallen to a generic ε -scale, not the problem-specific constants ρ_r^*, μ_r that would let a sharper (but non-detectable-without-them) choice do better. At $t \geq T_*$,

$$|u(t)| \leq \varepsilon^{1/L}|\log \varepsilon| + \frac{C_R}{c} \varepsilon^{(2L-1)/L} = O(\varepsilon^{1/L}|\log \varepsilon|),$$

since $\varepsilon^{(2L-1)/L} = o(\varepsilon^{1/L}|\log \varepsilon|)$ for $L \geq 2$. Finally, the factorisation $a^L - b^L = (a-b) \sum_{k=0}^{L-1} a^k b^{L-1-k}$ with $a = \sigma_r(T)$, $b = (\rho_r^*)^{1/L}$ bounds the sum by $L \max(a, b)^{L-1} = O(1)$ once $\sigma_r(T)$ is within a fixed neighbourhood of $(\rho_r^*)^{1/L}$, giving $|\hat{\rho}_r^{\text{plateau}}(\varepsilon, T) - \rho_r^*| = |\sigma_r(T)^L - \rho_r^*| \leq L \max(a, b)^{L-1} \cdot |u(T)| = O(\varepsilon^{1/L}|\log \varepsilon|)$ for all $T \geq T_*(\varepsilon)$, as claimed. $\square$

Lean status. The Lean proof of the plateau-tracking rate (`Saxe.signed_recovery_pos_magnitude_plateau`) uses qualitative convergence plus a topological existence argument for T_* , not the quantitative Grönwall bound above; both give the same asymptotic rate.

3 Negative-branch λ_r^* -rate (prop:neg-lambda)

Proposition 3.1 (Negative-branch λ_r^* -rate). *For $\rho_r^* < 0$, there exists an estimator $\hat{\lambda}_r^{\text{neg}}(\varepsilon, T)$, computable from σ_r on a window $[T/2, T]$ alone, achieving $|\hat{\lambda}_r^{\text{neg}} - |\lambda_r^*|| \leq C_\lambda^{\text{neg}} \varepsilon^{1/L}$ for ε small and T in a trajectory-detectable window, with C_λ^{neg} depending on L , μ_r , and C_R .*

Proof. Write $v(t) := \sigma_r(t)^{-(L-1)/L}$. By the chain rule and (1),

$$\dot{v}(t) = -\frac{L-1}{L} \sigma_r(t)^{-(2L-1)/L} \dot{\sigma}_r(t) = (L-1)|\lambda_r^*| + (L-1)\mu_r \sigma_r(t)^L - \frac{L-1}{L} \sigma_r(t)^{-(2L-1)/L} R_r(t, \varepsilon),$$

an *exact* identity (the leading $\sigma_r^{2-1/L}$ and $\sigma_r^{-(2L-1)/L}$ powers cancel exactly, leaving a constant plus two lower-order corrections). Integrating over $[T/2, T]$ and dividing by $L-1$ times the window length motivates the estimator

$$\hat{\lambda}_r^{\text{neg}}(\varepsilon, T) := \frac{\sigma_r(T)^{-(L-1)/L} - \sigma_r(T/2)^{-(L-1)/L}}{(L-1)(T/2)},$$

computable from σ_r on $[T/2, T]$ alone. Subtracting $|\lambda_r^*|$ and bounding each correction term using σ_r monotone decreasing on the negative branch (the paper's Theorem, sign trichotomy, case (iii)) gives

$$|\hat{\lambda}_r^{\text{neg}} - |\lambda_r^*|| \leq \mu_r \sigma_r(T/2)^L + \frac{C_R}{L} \varepsilon^{(2L-1)/L} \sigma_r(T)^{-(2L-1)/L}.$$

Take T any time by which σ_r has settled to its trajectory-regularity floor, $\sigma_r(t) = \Theta(\varepsilon^{1/L})$ for all t from then on — trajectory-detectable, since it only requires observing that σ_r has stopped decaying appreciably, not knowing λ_r^*, μ_r . Then the first term is $\Theta(\mu_r \varepsilon)$, and the second is $\Theta(\varepsilon^{(2L-1)(L-1)/L^2})$; both exponents exceed $1/L$ for every $L \geq 2$ (the second because $2(L-1)^2 \geq 2 > 1$), so both terms are $O(\varepsilon^{1/L})$, giving the claimed bound. $\square$

Why argued directly rather than cited. The Lean theorems for the negative branch (`SignedODE.sigma_negative_branch`, `SignedRecovery.signed_recovery_neg_lambda_rate`) assume the pre-correction ODE form (a deprecated, inverted-bracket equation superseded at companion-repository session 90/99), not (1) as this paper states it, and have not been re-verified against it. The main-text derivation above is therefore argued directly from (1) rather than cited to either Lean theorem.

4 End-to-end finite-sample rate (cor:finite-sample-end)

Corollary 4.1 (End-to-end finite-sample rate). *On an event of probability at least $1 - \nu$ under which $\|\hat{\Sigma}^{xx} - \Sigma^{xx}\|_{\text{op}}, \|\hat{\Sigma}^{yx} - \Sigma^{yx}\|_{\text{op}} \leq C'' \sqrt{\log(d/\nu)/n}$ (matrix-Bernstein concentration), the signed-decomposition estimator $\hat{\rho}_r$ computed in the sample eigenbasis $\{\hat{v}_r\}$ satisfies, uniformly in r ,*

$$|\hat{\rho}_r - \rho_r^*| \leq C \cdot \varepsilon^{1/L} |\log \varepsilon| + \frac{C'}{\delta} \sqrt{\frac{\log(d/\nu)}{n}},$$

with C, C' depending on L , the population covariances, and σ_x, σ_y .

Proof. Write $\hat{\rho}_r(v_r)$ for the estimator computed by reading σ_r off in the coordinate direction v_r ; the population estimator is $\hat{\rho}_r(v_r^*)$ and the actual computed one is $\hat{\rho}_r(\hat{v}_r)$. By the triangle inequality,

$$|\hat{\rho}_r(\hat{v}_r) - \rho_r^*| \leq \underbrace{|\hat{\rho}_r(\hat{v}_r) - \hat{\rho}_r(v_r^*)|}_{\text{basis perturbation}} + \underbrace{|\hat{\rho}_r(v_r^*) - \rho_r^*|}_{\text{dynamics error}}.$$

The second term is exactly the signed-decomposition theorem's bound, $O(\varepsilon^{1/L} |\log \varepsilon|)$ uniformly in r . For the first term, $v_r \mapsto \hat{\rho}_r(v_r)$ is Lipschitz on a neighbourhood of v_r^* with a constant depending only on $\|\bar{W}(t)\|$ along the trajectory (bounded, since $\bar{W}(t)$ converges to a fixed limit on each branch), so $|\hat{\rho}_r(\hat{v}_r) - \hat{\rho}_r(v_r^*)| \leq L_\rho \sin \angle(\hat{v}_r, v_r^*)$. Matrix Bernstein concentration [2, Thm 1.6.2] bounds the covariance perturbation $\|\hat{\Sigma}^{xx} - \Sigma^{xx}\|_{\text{op}}, \|\hat{\Sigma}^{yx} - \Sigma^{yx}\|_{\text{op}}$ at rate $O_p(\sqrt{\log(d/\nu)/n})$ with probability $1 - \nu$; a Wedin-type bound for the generalised eigenproblem converts this into $\sin \angle(\hat{v}_r, v_r^*) \leq (C_W/\delta) \sqrt{\log(d/\nu)/n}$, modulating the spectral gap δ . Combining the two terms gives the stated bound, with $C' := L_\rho C_W$. $\square$

5 Sign trichotomy (thm:trichotomy)

Theorem 5.1 (Sign trichotomy). *Under (1) with the bracket residual absorbed into R_r :*

- (i) *If $\rho_r^* > 0$, then $\sigma_r(t) \rightarrow (\rho_r^*)^{1/L}$ as $t \rightarrow \infty$, monotonically once $\sigma_r(t) > 0$.*
- (ii) *If $\rho_r^* = 0$, then $|\sigma_r(t)| = O(\varepsilon^{1/L})$ for all $t \in [0, t_{\max}^*]$; the bracket is identically $-\sigma^L$, damping any positive growth at the natural ε -scale.*
- (iii) *If $\rho_r^* < 0$, then $\sigma_r(t) \rightarrow 0$ monotonically; the bracket $-\sigma^L + \rho_r^* < 0$ for all $\sigma > 0$ forces strict decay.*

The constants in the convergence rates depend on L , ρ_r^ , μ_r , and the residual constant C_R .*

Proof. Proposition 1's residual bound is only asserted on the finite horizon $[0, t_{\max}^*]$, so the asymptotic ($t \rightarrow \infty$) claims below concern the idealised reduction $\dot{\sigma}_r = L\mu_r\sigma_r^{2-1/L}(\rho_r^* - \sigma_r^L)$ obtained by dropping R_r . The residual-aware, finite-time counterparts are §2 (case (i)) and §3 (case (iii)); case (ii)'s own gap at this idealisation is the main paper's Remark (rem:zero-ceiling-gap), unaffected by this relocation.

(i) The bracket $\rho_r^* - \sigma_r^L$ is positive for $\sigma_r < (\rho_r^*)^{1/L}$ and negative for $\sigma_r > (\rho_r^*)^{1/L}$, so for $\sigma_r > 0$, $\dot{\sigma}_r$ has the sign of $(\rho_r^*)^{1/L} - \sigma_r$: σ_r increases monotonically toward the fixed point from below and decreases toward it from above, so $\sigma_r(t) \rightarrow (\rho_r^*)^{1/L}$.

(ii) The bracket is identically $-\sigma_r^L$, so the idealised drift $-L\mu_r\sigma_r^{(3L-2)/L}$ is non-positive for $\sigma_r \geq 0$: whether this genuinely confines σ_r to the $O(\varepsilon^{1/L})$ scale against the residual R_r is exactly the content of the main paper's zero-ceiling-gap Remark, not re-derived here.

(iii) For $\rho_r^* < 0$ and $\sigma_r > 0$, the bracket $\rho_r^* - \sigma_r^L < 0$ always, so the idealised drift is strictly negative throughout, giving monotone decay. This case does not share case (ii)'s $L = 2$ pathology: at the initial scale $\sigma_r = \Theta(\varepsilon^{1/L})$ the drift magnitude is $\Theta(\varepsilon^{(2L-1)/L^2})$, which dominates the residual bound $C_R \varepsilon^{(2L-1)/L}$ for ε small since $(2L-1)/L^2 < (2L-1)/L$ for every $L \geq 2$ — unlike case (ii), where the idealised baseline drift is exactly zero and so cannot dominate any nonzero residual, here the baseline drift is itself nonzero and shrinks more slowly in ε than the residual, so strict decay survives the residual uniformly in L . $\square$

References

- [1] David Goh. Signed decomposition of the regression structure via linear JEPA training: An algorithm, empirical validation, and a machine-checked proof. AIS 2026, camera-ready.
- [2] Joel A. Tropp. An introduction to matrix concentration inequalities. *Foundations and Trends in Machine Learning*, 8(1-2):1–230, 2015.